

Question ID: 5127fc1e

Question

The north celestial pole (NCP)—the fixed point around which stars in the Northern Hemisphere (including the Sun) appear to rotate—is discernible only at night. Inspired by the navigational strategies of some insects and birds, researchers devised a method for locating the NCP in daytime using skylight polarization, which occurs as atmospheric particles scatter sunlight. A polarimetric camera captures images of polarization patterns, which rotate as the Sun’s position in the sky changes; temporal variances across images can then be used to determine an observer’s latitude and bearing relative to the NCP.

Which choice best describes the overall structure of the text?

Answer

- A. It illustrates how most navigational tools utilize the NCP, recounts how researchers discovered that certain animals are able to navigate without using the NCP, and then proposes that this discovery could be used to avoid problems in navigation associated with reliance on the NCP.
- B. It presents a celestial-based method of navigation, enumerates the comparative benefits of an alternative method used by certain animals that is based on an unrelated natural occurrence, and then indicates how researchers assessed the relative accuracy of the two methods.
- C. It explains how the NCP is typically located, emphasizes a key difference between how humans and certain animals use the NCP for navigation, and then suggests an alternative way of using the NCP to improve existing navigational instruments.
- D. It notes an obstacle to observing an astronomical phenomenon, mentions a navigational ability of certain animals that inspired a solution to that obstacle, and then explains how researchers used an optical device to mimic that ability.